

Atrial Fibrillation — Clinical Reference Summary

Diagnosis, Stroke Prevention, Rate/Rhythm Control

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1. Definition and Classification

Atrial fibrillation is a supraventricular tachyarrhythmia characterized by uncoordinated atrial electrical activation and, consequently, ineffective atrial contraction.

Pattern	Definition
Paroxysmal	Terminates spontaneously or with intervention within 7 days
Persistent	Continuous, lasting more than 7 days
Long-standing persistent	Continuous for more than 12 months
Permanent	Accepted, with no further attempts at rhythm control

2. The 'CC to ABC' Structured Approach (General Concept)

A widely referenced structured pathway for AF management proceeds through: Confirm the diagnosis, Characterize the arrhythmia (pattern, burden, underlying cause), and then follow an ABC pathway:

- A — Anticoagulation/Avoid stroke.
- B — Better symptom management (rate or rhythm control).
- C — Cardiovascular and comorbidity risk-factor optimization.

3. Stroke Risk Assessment

Stroke risk is commonly estimated using the CHA2DS2-VASc score, summarized below.

Criterion	Points
Congestive heart failure / LV dysfunction	1
Hypertension	1
Age ≥ 75 years	2
Diabetes mellitus	1
Prior stroke/TIA/thromboembolism	2
Vascular disease	1

Age 65–74 years	1
Sex category (female)	1

A score of 0 in men (or 1 in women, reflecting the sex-category point alone) is generally considered low risk, with anticoagulation typically not recommended; a score of ≥ 2 in men (or ≥ 3 in women) generally favors anticoagulation, with individualized decision-making for intermediate scores.

4. Anticoagulation

Direct oral anticoagulants are generally preferred over vitamin K antagonists for eligible patients with non-valvular atrial fibrillation, given comparable or superior efficacy with a generally lower risk of intracranial hemorrhage. Vitamin K antagonists remain preferred in patients with mechanical heart valves or moderate-to-severe mitral stenosis.

4.1 Bleeding Risk Assessment

Bleeding-risk scores (such as HAS-BLED) are used to identify and address modifiable bleeding risk factors — for example, uncontrolled hypertension, labile INR, or concurrent antiplatelet/NSAID use — rather than to exclude anticoagulation outright.

5. Rate Control

Beta-blockers and non-dihydropyridine calcium channel blockers are generally first-line for ventricular rate control. Digoxin may be added, particularly in patients with concurrent heart failure with reduced ejection fraction, given its neutral effect on blood pressure. A lenient initial rate-control target (resting heart rate <110 bpm) is commonly used, with stricter targets considered if symptoms persist.

6. Rhythm Control

- Antiarrhythmic drug therapy (e.g., flecainide, propafenone in structurally normal hearts; amiodarone or dronedarone in structural heart disease, per drug-specific contraindications).
- Catheter ablation (pulmonary vein isolation), increasingly considered earlier in the disease course, particularly for symptomatic paroxysmal AF or AF associated with heart failure.
- Direct current cardioversion for acute rhythm restoration, with attention to stroke-risk timing (anticoagulation before and after, or exclusion of atrial thrombus by transesophageal echocardiography).

7. Risk Factor and Lifestyle Modification

- Weight loss and management of obstructive sleep apnea, both associated with reduced AF burden and recurrence after ablation.
- Alcohol reduction, given its association with AF recurrence.
- Optimization of hypertension, diabetes, and heart failure, which are all AF risk amplifiers.

- Regular, moderate physical activity, avoiding extremes of endurance training associated with increased AF risk in some studies.

8. Special Populations

AF discovered incidentally on implanted device monitoring (subclinical AF/atrial high-rate episodes) requires individualized decision-making regarding anticoagulation, generally weighing episode duration and burden against stroke-risk score. Perioperative and post-cardiac-surgery AF is usually managed as a distinct, often self-limited entity, with anticoagulation decisions individualized based on persistence and overall stroke risk.